

Database Management System

Unit 1

1. Database System Applications

Database systems real-life me data store aur manage karne ke liye use hote hai.

Examples: banking systems, railway reservations, e-commerce websites, and hospital records.

Ye fast access, data security aur accuracy provide karte hai.

Jaha bhi large data handle karna hota hai, waha DBMS use hota hai.

2. Purpose of Database Systems

Database systems ka purpose hai data ko efficiently store, organize aur access karna.

Ye data redundancy ko kam karta hai aur consistency maintain karta hai.

Multiple users ek saath data use kar sakte hai.

Example: ek hi student record sab jagah same rahe, duplicate na ho.

3. View of Data

DBMS me user ko pura database nahi dikhaya jata, sirf required data hi dikhaya jata hai.

Isse data security aur simplicity maintain hoti hai.

3 levels hote hai: Physical (storage), Logical (structure), View (user-specific data).

Example: student sirf apne marks dekhta hai, pura database nahi.

4. Data Abstraction (Instance and Schema)

Data abstraction ka matlab hai unnecessary details hide karna aur simple view dena.

Schema = database ka structure (tables, fields design).

Instance = kisi particular time par database ka actual data.

Example: table design fixed rehta hai (schema), par data change hota rehta hai (instance).

5. Data Models

Data model batata hai ki data kaise store, organise aur relate hota hai.

Ye database design ka blueprint hota hai.

Common types: Hierarchical, Network, Relational, ER model.

Aajkal sabse jyada Relational model use hota hai (tables form me data).

6. ER Model (Entity Relationship Model)

ER Model database design ka graphical way hai jisme entities aur unke relationships show kiye jate hai.

Entities = real-world objects (Student, Course)

Relationships = unke beech connection (enrols)

Ye model database banane se pehle planning aur understanding me help karta hai.

7. Relational Model

Relational model me data tables (relations) ke form me store hota hai.

Har table me rows (records) aur columns (attributes) hote hai.

Tables ek dusre se keys ke through connected hote hai.

Ye simple, flexible aur most widely used data model hai.

8. Database Languages (DDL, DML, DCL, TCL)

DBMS me data manage karne ke liye different languages use hoti hai:

DDL → structure create/modify (CREATE, ALTER)

DML → data insert/update/delete (INSERT, UPDATE, DELETE)

DCL → permissions (GRANT, REVOKE)

TCL → transactions control (COMMIT, ROLLBACK)

Ye sab SQL ka part hote hai.

9. Database Design

Database design ka matlab hai data ko proper structure me organise karna.

Isme tables, relationships aur constraints plan kiye jate hai.

Goal: redundancy kam ho, data consistent rahe aur performance achha ho.

Achha design future me data handling easy aur efficient bana deta hai.

10. ER Diagram

ER Diagram ek visual diagram hota hai jo entities, attributes aur relationships ko show karta hai.

Isme rectangles (entities), ovals (attributes), aur diamonds (relationships) use hote hai.

Ye database design ko easily samajhne aur plan karne me help karta hai.

11. Entities and Attributes

Entity koi real-world object hota hai (Student, Teacher).

Attributes us entity ke properties hote hai (name, age, id).

Example: Student entity ke attributes → roll_no, name, marks.

Entities ko uniquely identify karne ke liye key attribute use hota hai.

12. Entity Sets

Entity set same type ki entities ka collection hota hai.

Example: sabhi students ka group = Student entity set.

Har entity set me similar attributes hote hai.

Ye table jaisa hota hai jisme multiple records (entities) store hote hai.

13. Relationships and Relationship Sets

Relationship entities ke beech connection batata hai.

Example: Student "enrols in" Course.

Relationship set same type ke relationships ka collection hota hai.

Ye batata hai kaise different entity sets ek dusre se related hai.

14. ER Design Issues

ER design me kuch challenges aate hai jaise entity ya attribute decide karna, redundancy avoid karna aur proper relationships define karna.

Special cases: weak entities, multivalued attributes, specialisation/generalisation.

Goal: clear, simple, and accurate database structure design karna.

Unit 2

1. Introduction to Relational Model

Relational model me data tables (relations) ke form me represent hota hai.

Har table me rows (tuples) aur columns (attributes) hote hai.

Tables ek dusre se keys (primary/foreign) ke through connected hote hai.

Ye model simple, logical aur most widely used hai DBMS me.

2. Structure of Relational Database

Relational database tables (relations) se bana hota hai.

Har table me rows (tuples) aur columns (attributes) hote hai.

Har table ka ek unique name hota hai aur primary key se records identify hote hai.

Foreign key tables ke beech relationship banati hai.

3. Database Schema

Database schema database ka blueprint hota hai.

Ye batata hai kaunse tables honge, unke columns kya honge aur relationships kaise honge.

Schema fixed rehta hai, easily change nahi hota.

Example: Student table ka structure (id, name, marks) = schema.

4. Keys

Keys database me records ko uniquely identify karne ke liye use hoti hai.

Primary key → unique identification (roll_no)

Foreign key → tables ko connect karta hai

Candidate key → possible primary keys

Keys data consistency aur relationships maintain karne me important role play karti hai.

5. Schema Diagram

Schema diagram database ka visual representation hota hai jisme tables aur unke relationships dikhaye jate hai.

Isme table names, attributes aur keys show hote hai.

Ye developers ko structure samajhne aur database design plan karne me help karta hai.

6. Basic Structure of SQL Queries

SQL queries ka basic structure hota hai:

SELECT columns

FROM table

WHERE condition

SELECT data choose karta hai, FROM table batata hai, aur WHERE condition filter karta hai.

Example: student table se marks > 50 wale records nikalna.

7. Set Operations

Set operations multiple query results ko combine karne ke liye use hote hai.

Main types: UNION (combine + remove duplicates), UNION ALL (combine + allow duplicates), INTERSECT (common data), EXCEPT (difference).

Example: 2 tables ke common students nikalna.

8. Aggregate Functions

Aggregate functions multiple rows par calculation karke ek single result deti hai.

Common functions: COUNT(), SUM(), AVG(), MAX(), MIN()

Example: class ke total students ya average marks calculate karna.

Ye mostly GROUP BY ke saath use hoti hai data analysis ke liye.

9. GROUP BY and HAVING

GROUP BY similar data ko groups me divide karta hai.

HAVING group par condition apply karta hai (WHERE row par lagta hai).

Example: har subject ka average marks nikalna (GROUP BY subject) aur sirf >60 wale groups show karna (HAVING).

10. Nested Subqueries

Nested subquery ek query ke andar dusri query hoti hai.

Inner query pehle execute hoti hai, uska result outer query use karti hai.

Example: highest marks wale student ko find karna using subquery.

Ye complex conditions handle karne me useful hoti hai.

11. Joins

Joins multiple tables ko combine karne ke liye use hote hai based on related columns.

Types: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN

Example: Student table aur Course table ko join karke student ke enrolled courses dikhana.

Ye relational database ka core concept hai.

12. Triggers

Trigger ek automatic action hota hai jo database me kisi event par execute hota hai (INSERT, UPDATE, DELETE).

Ye predefined rules follow karta hai.

Example: jab new student add ho, tab automatic log entry create ho jaye.

Triggers data integrity aur automation ke liye use hote hai.